

Forecasting Corporate Credit Spreads Using Macroeconomic and Financial Indicators

An Expanding-Window Regression Framework for One-Month-Ahead Forecasting

Capstone Project

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1. Introduction

Corporate credit spreads reflect the additional yield investors require to hold corporate debt relative to lower-risk government securities. Changes in these spreads provide information about perceived credit risk, economic conditions, market uncertainty, and investor risk appetite. Widening spreads are generally associated with deteriorating credit conditions or increased market stress, while tightening spreads often accompany improving economic conditions and stronger investor confidence.

Because credit spreads respond to both macroeconomic developments and financial-market conditions, forecasting their movements is relevant to investment management, credit-risk analysis, and portfolio positioning. However, their behavior can be difficult to predict because the relationships between spreads and economic or market variables may change over time and across market regimes.

This project investigates whether macroeconomic, interest-rate, and market information can be used to forecast changes in corporate credit spreads one month ahead

2. Project Objective

The objective of this project was to develop and evaluate a forecasting framework for the Moody's Baa corporate bond yield spread over the 10-year U.S. Treasury yield. The model was designed to predict the credit spread one month ahead using information that would have been available at the time the forecast was made.

The analysis incorporated macroeconomic indicators, interest-rate variables, financial-market measures, and lagged characteristics of the credit spread itself. Particular attention was given to the timing of economic data releases and the alignment of variables to reduce look-ahead bias and create a more realistic forecasting framework.

Several forecasting approaches were evaluated against simple benchmarks, with performance assessed using both forecast-error measures and directional accuracy. In addition to predicting the level of the future spread, an important objective was to determine whether the model could meaningfully identify whether credit conditions were likely to widen or tighten during the following month.

3. Business Context

Credit spreads are an important measure of risk conditions in corporate debt markets and can influence portfolio construction, risk management, and investment decisions. Portfolio managers and quantitative researchers may use expectations about future spread movements when evaluating corporate-bond exposure, while credit-risk and risk-management teams can use changes in spreads as an indicator of shifting market perceptions of credit risk.

A model capable of identifying an increased probability of spread widening could provide an additional quantitative signal of deteriorating credit conditions, while expectations of spread tightening may indicate improving conditions or greater investor willingness to take credit risk. Such forecasts are not intended to replace fundamental credit analysis, but rather to complement it with a systematic, data-driven assessment of expected market conditions.

4. Intended Audience

This project is intended for professionals interested in quantitative research, financial markets, and data-driven investment analysis. By examining the relationship between macroeconomic conditions, market indicators, and corporate credit spreads, the analysis may be particularly relevant to:

- Quantitative Researchers and Analysts
- Fixed-Income Portfolio Managers

- Asset Management and Investment Professionals
- Macro and Market Research Analysts
- Credit Risk Analysts
- Treasury and Balance-Sheet Risk Teams
- Data Scientists working in financial markets

5. Data Source

Unlike projects based on a pre-existing modeling dataset, the analytical dataset used in this project was constructed specifically for the forecasting objective. Individual macroeconomic, interest-rate, credit-market, and financial-market time series were selected based on their potential relationship with corporate credit spreads and collected separately from multiple sources, primarily Federal Reserve Economic Data (FRED), Archival Federal Reserve Economic Data (ALFRED), and Yahoo Finance. The selected series include the Baa–10Y corporate credit spread, Consumer Price Index (CPI), unemployment rate, industrial production, federal funds rate, real GDP, 10-year minus 3-month Treasury yield spread, 10-year Treasury yield, VIX, S&P 500, Russell 2000, and WTI crude oil prices.

The individual series were retrieved using a combination of direct data downloads and API access. FRED and ALFRED data were used for macroeconomic, interest-rate, credit-spread, volatility, and oil-price series, while historical S&P 500 and Russell 2000 market data were obtained from Yahoo Finance. For selected macroeconomic indicators, including industrial production and GDP, historical observations were combined with information on their initial public-release timing to better represent when the data would actually have been available to a forecaster.

Data Series and Sources			
Data Series	Source	Retrieval Method	Release treatment
Baa–10Y Corporate Credit Spread	FRED	Direct download	Historical series data
Consumer Price Index (CPI)	FRED / ALFRED	API	Initial-release / vintage data
Unemployment Rate	FRED / ALFRED	API	Initial-release / vintage data
Industrial Production Index	FRED / ALFRED	API	Historical data aligned to initial-release timing
Federal Funds Rate	FRED	Direct download	Historical series data

Data Series and Sources			
Real GDP	FRED / ALFRED	API	Historical data aligned to initial-release timing
10-Year Minus 3-Month Treasury Yield Spread	FRED	Direct download	Historical series data
10-Year Treasury Yield	FRED	Direct download	Historical series data
VIX	FRED	Direct download	Historical series data
S&P 500	Yahoo Finance	Yahoo Finance API / Python	Historical market data
Russell 2000	Yahoo Finance	Yahoo Finance API / Python	Historical market data
WTI Crude Oil Price	FRED	Direct download	Historical series data

The collected series were subsequently cleaned, transformed where appropriate, and integrated into a monthly analytical dataset. Because macroeconomic indicators are released at different times and may also be revised after their initial publication, particular attention was given to the historical availability of information to reduce look-ahead bias. The detailed data transformations, release-date treatment, and temporal-alignment methodology are discussed in the following sections.

6. Data Preparation, Feature Engineering, and Temporal Alignment

After the individual economic and financial time series were collected, they were cleaned, transformed, and integrated into a consistent monthly dataset for forecasting. Because the source variables differed in frequency, publication timing, and scale, the preparation process included both variable-specific transformations and temporal alignment. Particular attention was given to ensuring that predictor values reflected information that would have been available at the time each forecast was made.

Several variables were transformed to better represent changes in economic and market conditions. Macroeconomic series were converted to year-over-year growth measures where appropriate, financial-market price series were converted to monthly returns, and a rolling measure of credit-spread volatility was constructed. The forecasting target was also shifted forward by one month to establish the one-month-ahead prediction framework.

6.1 Forecast Target Definition

The forecasting target was constructed from the Moody's Baa corporate bond yield spread over the 10-year U.S. Treasury yield. The objective was to predict the level of the corporate credit spread one month ahead using information available in the current and prior periods.

To create the target variable, the credit-spread series was shifted backward by one observation so that the spread observed in month $t+1$ was assigned as the target for the predictor information associated with month t . The resulting target can therefore be expressed as:

$$y_t = \text{Credit Spread}_{t+1}$$

Under this structure, each row of the modeling dataset represents the information available for forming a forecast at time t , while the corresponding target represents the corporate credit spread observed in the following month. This transformation established the one-month-ahead forecasting horizon used consistently throughout model development and evaluation.

6.2 Macroeconomic Transformations

Three macroeconomic variables were transformed into year-over-year measures: inflation, industrial production growth, and real GDP growth. These transformations were used to represent the pace of change in underlying economic conditions rather than the absolute levels of the series. While year-over-year measures are less sensitive to short-term fluctuations, they provide interpretable measures of the broader macroeconomic environment that complement the higher-frequency market and credit variables included in the forecasting framework.

For inflation, CPI observations were obtained using ALFRED vintage data, which provided both the reported values and their historical release information. Year-over-year inflation was calculated as the percentage change in CPI relative to the same month of the previous year. This allowed the inflation measure to be constructed using observations associated with their historical release dates.

Industrial production required a different approach. Historical observations were combined with initial-release timing information obtained from ALFRED. The historical series was used to calculate year-over-year industrial production growth, while the release information was used to determine when each observation would have become available. This approach was adopted because benchmark revisions in the vintage series produced inconsistencies and unusually large growth observations for

some historical periods. Using the historical series for the growth calculation while retaining the initial-release timing provided a more stable measure while preserving the historical availability of the information as closely as possible.

A similar procedure was applied to real GDP. Because GDP is reported quarterly, year-over-year growth was calculated by comparing each quarterly observation with the corresponding quarter of the previous year. Historical GDP observations were used for the growth calculation, while ALFRED release information was used to identify when the corresponding GDP information became publicly available. During subsequent monthly dataset alignment, each available quarterly GDP growth observation was carried forward until a new GDP observation became available, allowing the most recently available GDP information to be used for each monthly forecast.

6.3 Financial-Market Transformations

The S&P 500, Russell 2000, and WTI crude oil series were transformed from higher-frequency market data into monthly return measures. For each series, the last available observation within each calendar month was retained to represent the month-end value. This approach provided a consistent monthly frequency for integration with the broader forecasting dataset.

Monthly returns were used instead of price or index levels to capture recent changes in financial-market conditions. Whereas the absolute level of an equity index or commodity price may reflect a long-term trend, monthly percentage changes provide a more direct measure of recent market performance and allow movements across series with very different scales to be compared more consistently.

For the S&P 500 and Russell 2000, historical market data obtained from Yahoo Finance were reduced to month-end closing values. Monthly returns were then calculated as the percentage change between the closing value on the last available trading day of the current month and the corresponding month-end value from the previous month:

$$R_t = (P_t - P_{t-1}) / P_{t-1}$$

where P_t represents the month-end closing value in month t and P_{t-1} represents the month-end closing value in the preceding month.

WTI crude oil prices obtained from FRED were processed using the same monthly aggregation approach. The last available daily oil-price observation within each month was retained, and monthly oil returns were calculated from consecutive month-end values. Converting the three market series to monthly returns

allowed their movements to be compared on a consistent basis and incorporated into the monthly forecasting framework.

6.4 Six-Month Rolling Credit Spread Volatility

A six-month rolling measure of credit-spread volatility was constructed as an additional predictor to capture the recent stability of credit-market conditions. While the level of the credit spread provides information about prevailing credit conditions, the degree to which spreads have fluctuated over recent months may provide additional information about changes in market uncertainty and risk.

The volatility measure was calculated as the rolling standard deviation of the corporate credit spread over the current and previous five months:

$$S_{t,6} = \text{SD}(S_t, S_{t-1}, \dots, S_{t-5})$$

where S_t represents the corporate credit spread in month t . A six-month window was selected rather than a longer window, such as twelve months, to place greater emphasis on relatively recent credit-market behavior and allow the measure to respond more quickly to changing market conditions.

Higher values of the resulting feature indicate greater recent variation in credit spreads, while lower values indicate more stable spread behavior.

6.5 Financial-Market and Interest-Rate Data Preparation

Financial-market and interest-rate variables were converted to a common monthly frequency using aggregation methods selected according to the economic interpretation of each series. Rather than applying a single aggregation method to all daily variables, month-end observations or monthly averages were used depending on whether the objective was to capture conditions prevailing at the end of the forecasting month or conditions experienced throughout the month.

For the S&P 500, Russell 2000, and WTI crude oil series, the last available observation within each calendar month was retained. Monthly returns were subsequently calculated from consecutive month-end values. Using month-end observations provided a measure of the change in market conditions between successive forecasting dates, while converting the series to returns allowed changes across variables with different scales to be represented more consistently.

The 10-year U.S. Treasury yield and the 10-year minus 3-month Treasury yield spread were also derived from daily data using the last available observation of each month. Unlike the equity and oil series, however, these variables were retained in levels rather than converted to returns. The month-end 10-year Treasury yield represents the prevailing long-term interest-rate environment at the forecasting date, while the 10-year minus 3-month spread represents the slope of the yield curve. Retaining the final available monthly observation was intended to reflect the most recent interest-rate information available before forecasting the following month's credit spread.

VIX was treated differently because the objective was to capture the overall level of market uncertainty experienced during the month rather than conditions on a single date. Daily VIX observations were therefore averaged within each month. A monthly average reduces the dependence of the measure on the particular month-end observation and preserves information about elevated volatility that may have occurred earlier in the month.

The Federal Funds Rate series was already available at a monthly frequency and therefore did not require daily-to-monthly aggregation. The rate was retained in level form because its level directly represents the prevailing short-term interest-rate environment.

6.5 Dataset Integration and Temporal Alignment

Following the individual data transformations, all variables were integrated into a common monthly dataset. The forecasting framework was designed so that information available during month t was used to predict the corporate credit spread in month $t+1$. Because the source variables differed in frequency and publication schedule, all observations were ultimately assigned to a common month-end index based on when the information was available. The month-end date therefore represents the information set accumulated throughout that month rather than implying that every variable was observed specifically on the final calendar day.

Release timing was particularly important for macroeconomic indicators. For example, the unemployment rate for a given reference month is generally released during the following month. The historical release date was therefore used to determine when each unemployment observation became available to the forecasting model. Once released, the observation was assigned to the month in which that information became known and subsequently aligned to that month's common month-end index. This prevented an unemployment observation from being used in a forecast before it would actually have been available.

Missing-Value Treatment

The treatment of missing observations depended on their source and economic interpretation. Missing values in daily financial-market and interest-rate series were generally associated with weekends, holidays, or other non-trading days and were not artificially filled. For variables represented by month-end values, the last available observation within the month was retained. For VIX, available daily observations were used to calculate the monthly average, so isolated non-trading-day observations did not require imputation.

Missing macroeconomic observations associated with delayed data releases, including periods affected by government shutdowns, were treated differently. When a new observation was not yet available, the previously available value was carried forward. This approach reflects the information that would actually have been available to a forecaster at that point in time rather than introducing subsequently released information into an earlier forecasting period.

Quarterly real GDP growth was similarly carried forward across monthly observations until new GDP information became available. This allowed the monthly dataset to reflect the most recently available GDP information without assigning future information to months in which it had not yet been released.

After these adjustments, all series were indexed to a common month-end timeline and merged into the final analytical dataset. Each row therefore represents the information available through a given month, while the shifted target represents the corporate credit spread in the subsequent month. This alignment established a consistent one-month-ahead forecasting framework and was designed to minimize look-ahead bias throughout model development and evaluation.

7. Exploratory Data Analysis

Before constructing the forecasting models, exploratory data analysis was performed to examine the behavior and distribution of the credit spread and the macroeconomic and financial variables included in the dataset. The analysis focused on identifying differences in variable distributions, changes in credit-market conditions over time, persistence in the credit-spread series, relationships among predictors, and potential lagged relationships relevant to one-month-ahead forecasting.

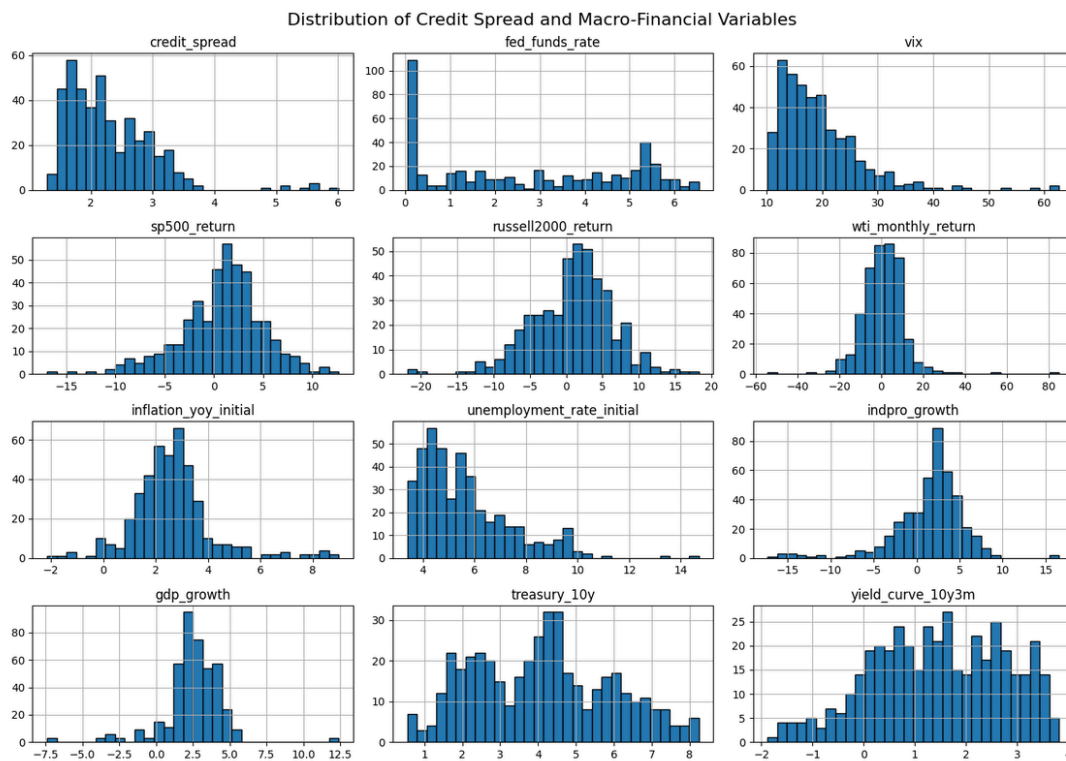
7.1 Descriptive Analysis

The final integrated dataset contains 427 monthly observations across a broad range of economic and financial conditions. The Baa–Treasury credit spread averaged approximately 2.29 percentage points, with observations ranging from 1.29% to 6.01%. The distribution is right-skewed, with most observations concentrated at relatively moderate spread levels and a smaller number of substantially higher

observations corresponding to periods of elevated credit-market stress.

The explanatory variables also exhibit substantially different distributions and degrees of variability. VIX and unemployment are right-skewed, reflecting occasional periods of unusually high financial or economic stress. Inflation is concentrated primarily around moderate positive rates but includes both negative observations and periods of considerably higher inflation. Industrial production and real GDP growth are generally centered around positive growth rates but contain large negative observations associated with periods of economic contraction.

Financial-market returns are centered closer to zero and contain both positive and negative observations. Russell 2000 returns display a wider distribution than S&P 500 returns, indicating greater variability in small-cap equity performance over the sample. WTI crude-oil returns exhibit particularly large fluctuations, reflecting the higher volatility of energy prices. The 10-year minus 3-month Treasury yield spread includes both positive and negative observations, allowing the dataset to capture periods of both positively sloped and inverted yield curves.

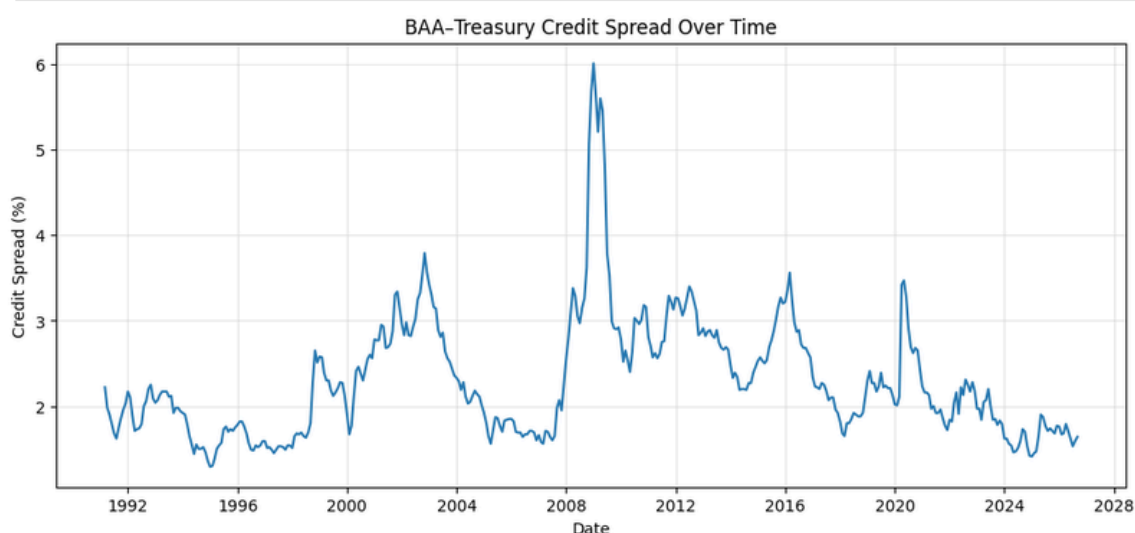


Overall, the descriptive analysis indicates that the dataset covers both relatively stable economic environments and periods of substantial financial and macroeconomic stress. The presence of skewness, extreme observations, and differences in variability across predictors also suggests that relationships with credit spreads may not remain constant across all market conditions.

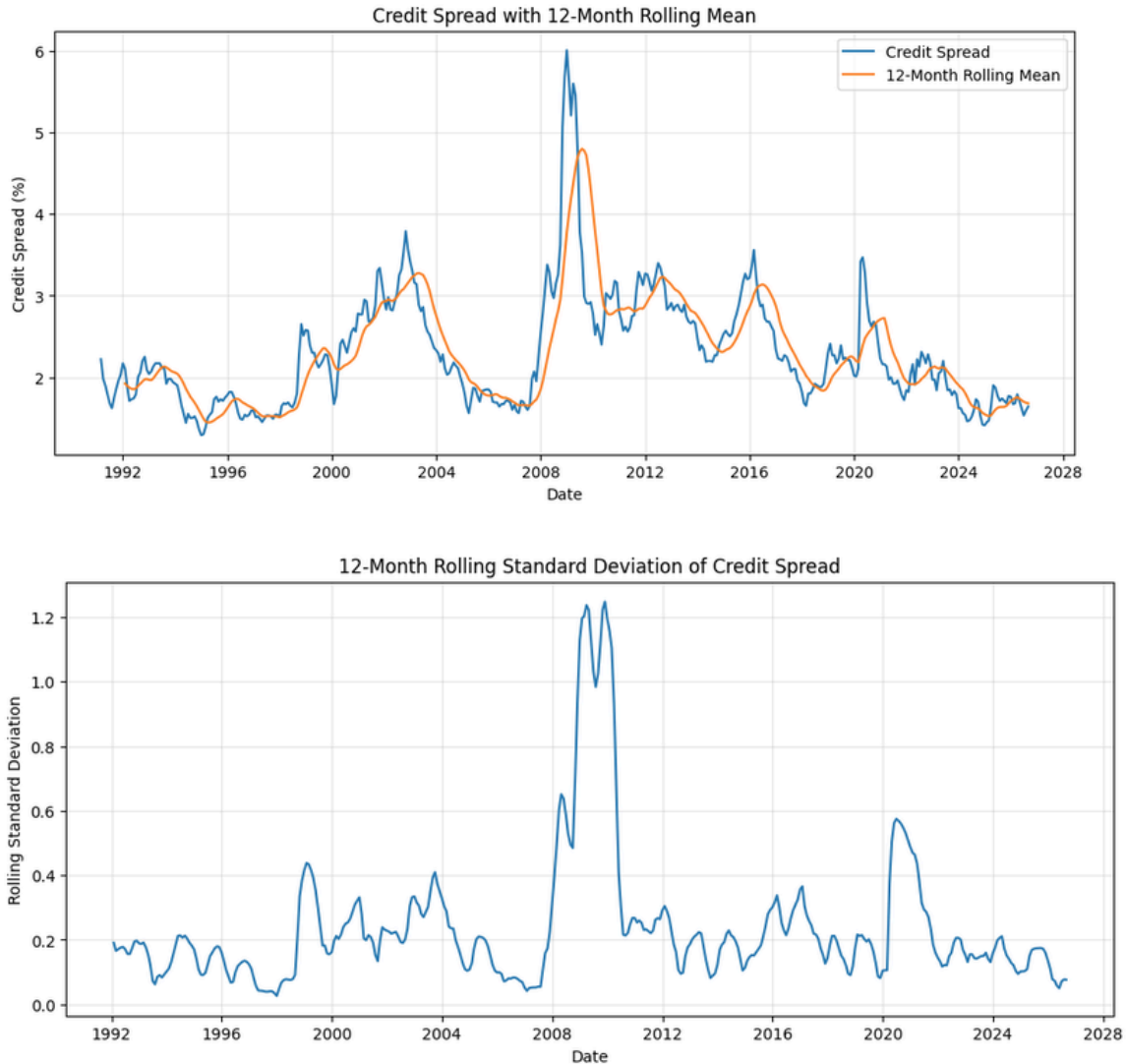
7.2 Time-Series Analysis

The historical behavior of the Baa–Treasury credit spread was examined to assess its persistence, stability, and behavior across different market environments. The analysis included the spread level over time, rolling measures of its mean and variability, a stationarity test, and its autocorrelation structure.

The credit-spread series exhibits distinct periods of widening associated with economic and financial stress. The most pronounced increase occurs during the 2008–2009 Global Financial Crisis, when the spread approached 6 percentage points. Other periods of elevated spreads are visible during the early-2000s downturn, the 2015–2016 period of market and commodity-related stress, and the 2020 COVID-19 shock. These episodes are followed by subsequent declines in spreads, indicating that periods of extreme widening have historically tended to be temporary rather than permanent.



Rolling statistics provide additional evidence that the behavior of the series changes across market environments. The 12-month rolling mean highlights persistent periods of elevated credit spreads, while the 12-month rolling standard deviation rises sharply during major stress episodes, particularly during the Global Financial Crisis and, to a lesser extent, around the COVID-19 shock. This indicates that although the spread tends to return toward more typical levels following periods of stress, its variability is not constant over time.



7.3 Stationarity Analysis

Before developing the forecasting models, the stationarity of the corporate credit-spread series was evaluated. Stationarity is an important consideration in time-series modeling because a nonstationary series may contain persistent trends or a unit root, which can produce unstable or potentially spurious relationships when the series is modeled in levels. Testing the target series therefore helped determine whether the credit spread could reasonably be modeled in its level form or whether additional transformations, such as differencing, might be necessary.

The **Augmented Dickey–Fuller (ADF)** test was applied to test for the presence of a unit root. The null hypothesis of the ADF test is that the series contains a unit root and is therefore nonstationary, while rejection of the null provides evidence in favor of stationarity.

The test produced an ADF statistic of approximately -3.77 and a p-value of 0.0032 . Because the p-value is below 0.01 , the null hypothesis of a unit root was rejected at the 1% significance level. The results therefore provide strong statistical evidence that the corporate credit-spread series is stationary in the unit-root sense.

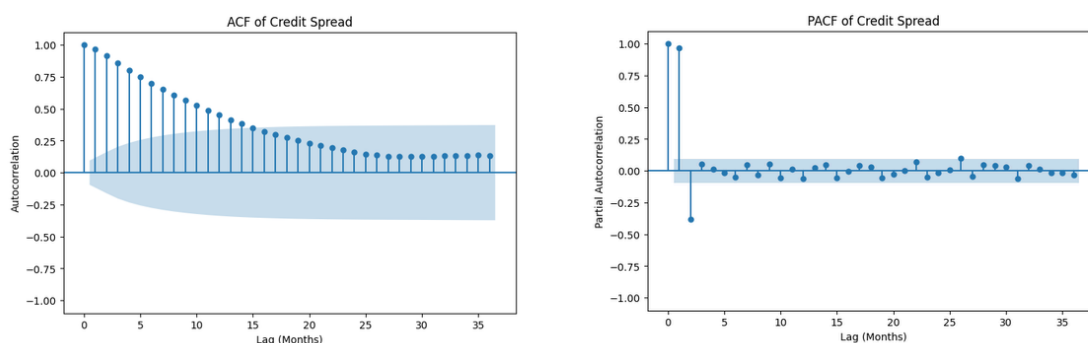
This finding supported the decision to model the credit spread in levels rather than applying differencing solely to achieve stationarity. However, stationarity does not imply that the variability of the series is constant over time. As observed in the preceding time-series analysis, credit-spread volatility increases substantially during periods of financial stress.

7.4 Autocorrelation Analysis

To further examine the time-series structure of the corporate credit spread, the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) were analyzed. These measures evaluate the relationship between the current value of a time series and its previous values and can help identify whether historical credit spreads contain useful information for forecasting future spreads.

The ACF measures the correlation between the current credit spread and its values at different previous lags. For example, the correlation at lag 1 measures the relationship between the current month's spread and the previous month's spread, while lag 2 measures the relationship between the current spread and the spread two months earlier. Importantly, the ACF includes both direct and indirect relationships across intermediate lags. Therefore, if the credit spread is highly persistent from one month to the next, that persistence can produce significant autocorrelation across several subsequent lags.

The PACF isolates the direct relationship between the current credit spread and a particular lag after controlling for the effects of the intervening lags. For example, the PACF at lag 2 measures the relationship between the current spread and the spread two months earlier after accounting for the information already contained in the one-month lag. This distinction makes the PACF particularly useful for determining whether additional lags contribute information beyond the most recent observations.



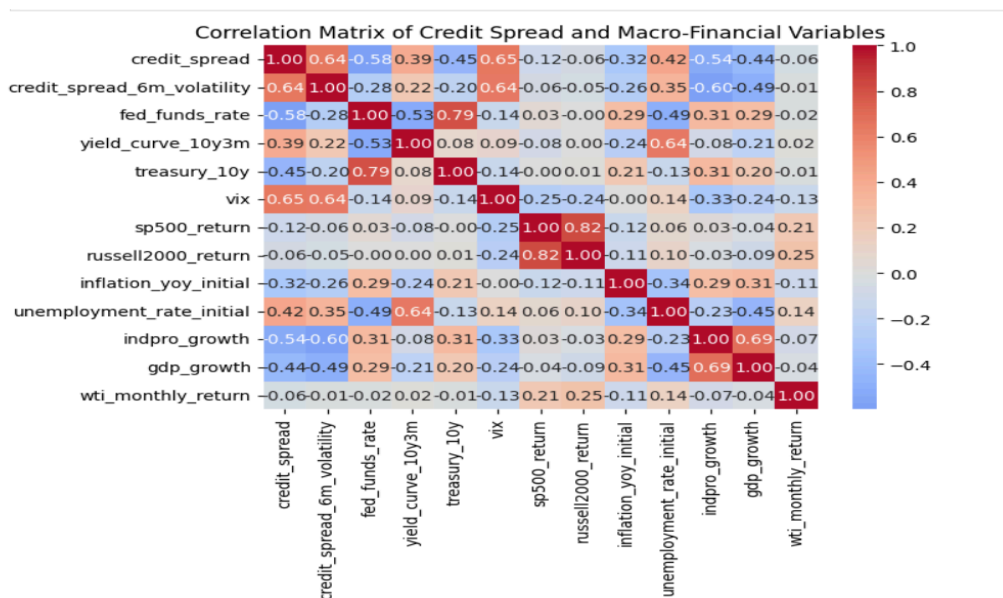
The ACF indicates strong positive autocorrelation at the first lag and persistent positive correlations across subsequent lags, demonstrating that corporate credit spreads are highly persistent over time. In other words, a high spread in one month tends to be followed by relatively high spreads in subsequent months, while lower-spread environments also tend to persist.

The PACF provides additional information about the source of this persistence. The first lag displays a particularly strong positive partial autocorrelation, indicating that the immediately preceding month's credit spread contains substantial information about the current spread. The second lag also provides additional information after controlling for the first lag, although its contribution is considerably smaller. Partial autocorrelations at subsequent lags are generally weaker, suggesting that much of the predictive information contained in the spread's recent history is concentrated in the earliest lags.

Together, the ACF and PACF results provide evidence of substantial short-term persistence in corporate credit spreads and support the inclusion of lagged credit-spread information in the forecasting framework. They also motivate the use of an autoregressive benchmark later in the analysis, allowing the predictive contribution of macroeconomic and financial-market variables to be evaluated relative to a model based primarily on the spread's own historical behavior.

7.5 Correlation Analysis

Correlation analysis was performed to examine the strength and direction of contemporaneous linear relationships between the corporate credit spread and the macroeconomic and financial variables, as well as relationships among the predictors themselves. This analysis provides an initial indication of which variables tend to move with credit spreads and helps identify potential multicollinearity among predictors.



Several economically meaningful relationships are visible in the correlation matrix. The strongest positive correlations with the credit spread are observed for VIX (approximately 0.65) and six-month credit-spread volatility (approximately 0.64). These relationships are consistent with wider credit spreads occurring during periods of elevated financial-market uncertainty and greater instability in credit conditions. Unemployment and the 10-year minus 3-month yield-curve spread also show moderate positive contemporaneous relationships with credit spreads.

Several macroeconomic and interest-rate variables exhibit negative correlations with the credit spread. The Federal Funds Rate has a correlation of approximately -0.58 , while industrial production growth, the 10-year Treasury yield, and real GDP growth also display moderate negative relationships. These correlations indicate that wider spreads in the historical sample have tended to coincide with weaker economic activity and particular interest-rate environments. However, these relationships are descriptive and should not be interpreted as evidence of causality.

Equity and oil returns exhibit relatively weak contemporaneous correlations with the level of the credit spread. This does not necessarily imply that these variables lack forecasting value, since the forecasting problem concerns the relationship between information available in one period and the credit spread in a subsequent period. Lagged relationships are therefore examined separately.

The correlation matrix also reveals substantial relationships among several predictors. S&P 500 and Russell 2000 monthly returns are highly correlated (approximately 0.82), while the Federal Funds Rate and 10-year Treasury yield are also strongly positively correlated (approximately 0.79). Industrial production and real GDP growth exhibit a correlation of approximately 0.69, and VIX is strongly related to the constructed credit-spread volatility measure (approximately 0.64). These relationships indicate potential multicollinearity and overlap in the information captured by some predictors.

Because the primary objective of the analysis is out-of-sample forecasting rather than interpretation of individual regression coefficients, correlated predictors were not automatically excluded based solely on pairwise correlation. Some correlated variables also represent economically distinct information; for example, the Federal Funds Rate reflects short-term monetary-policy and funding conditions, while the 10-year Treasury yield represents longer-term interest-rate conditions. The candidate predictors were therefore retained for subsequent forecasting analysis, while the presence of multicollinearity was considered when interpreting individual regression coefficients.

8. Lagged Feature Construction

As described in Section 6.1, the forecasting target was defined as the corporate credit spread one month ahead. Following the exploratory analysis, lagged versions of the predictor variables were constructed to examine whether historical macroeconomic, interest-rate, and financial-market conditions contained information about future credit spreads.

The lagged feature set included previous values of the credit spread itself as well as lagged values of the macroeconomic and market predictors. This was particularly relevant for the credit spread because the ACF and PACF analysis demonstrated substantial persistence in the series, suggesting that recent spread levels may contain important information about the following month's spread.

8.1 Predictor Lag Construction

Lagged features were constructed at 1-, 3-, 6-, and 12-month horizons, representing short-term, quarterly, semiannual, and annual historical information. These candidate horizons allowed the analysis to compare whether more recent observations or conditions further in the past exhibited stronger relationships with the one-month-ahead credit-spread target.

For each predictor X_t , the lagged variables were defined as:

$$X_{t-1}, X_{t-3}, X_{t-6}, X_{t-12}$$

The same lag structure was applied to the credit spread, six-month credit-spread volatility, federal funds rate, yield-curve slope, 10-year Treasury yield, VIX, S&P 500 returns, Russell 2000 returns, WTI crude-oil returns, industrial production growth, real GDP growth, inflation, and unemployment.

Using lagged variables also preserved the temporal structure of the forecasting problem: historical observations were used as potential predictors of a future credit spread rather than allowing future information to enter the predictor set.

8.2 Feature–Target Correlations

After constructing the lagged features, the correlation between each candidate feature and the one-month-ahead credit-spread target was calculated. The correlations were ranked by absolute magnitude, from strongest to weakest, while retaining their original signs. This approach allowed both strong positive and strong negative relationships to be identified, since either direction may contain useful predictive information.

The results demonstrated particularly strong persistence in the credit-spread series. The current credit spread had the strongest relationship with the one-month-ahead target (0.97), followed by the one-month lag (0.92) and three-month lag (0.81). The relationship declined at longer horizons, with the six-month spread lag having a correlation of 0.66 and the 12-month lag 0.42. This pattern was consistent with the earlier ACF/PACF analysis and indicated that recent credit-spread information was particularly relevant to near-term spread behavior.

Several other variables also showed notable relationships with the future spread. The contemporaneous VIX had a correlation of 0.66, while its one-month lag remained relatively strong at 0.62. Six-month credit-spread volatility had a correlation of 0.61. Among the macroeconomic and interest-rate variables, the federal funds rate (−0.55) and industrial production growth (−0.50) showed relatively strong negative relationships with the target, while the 10-year Treasury yield, GDP growth, yield-curve slope, and unemployment rate exhibited moderate relationships.

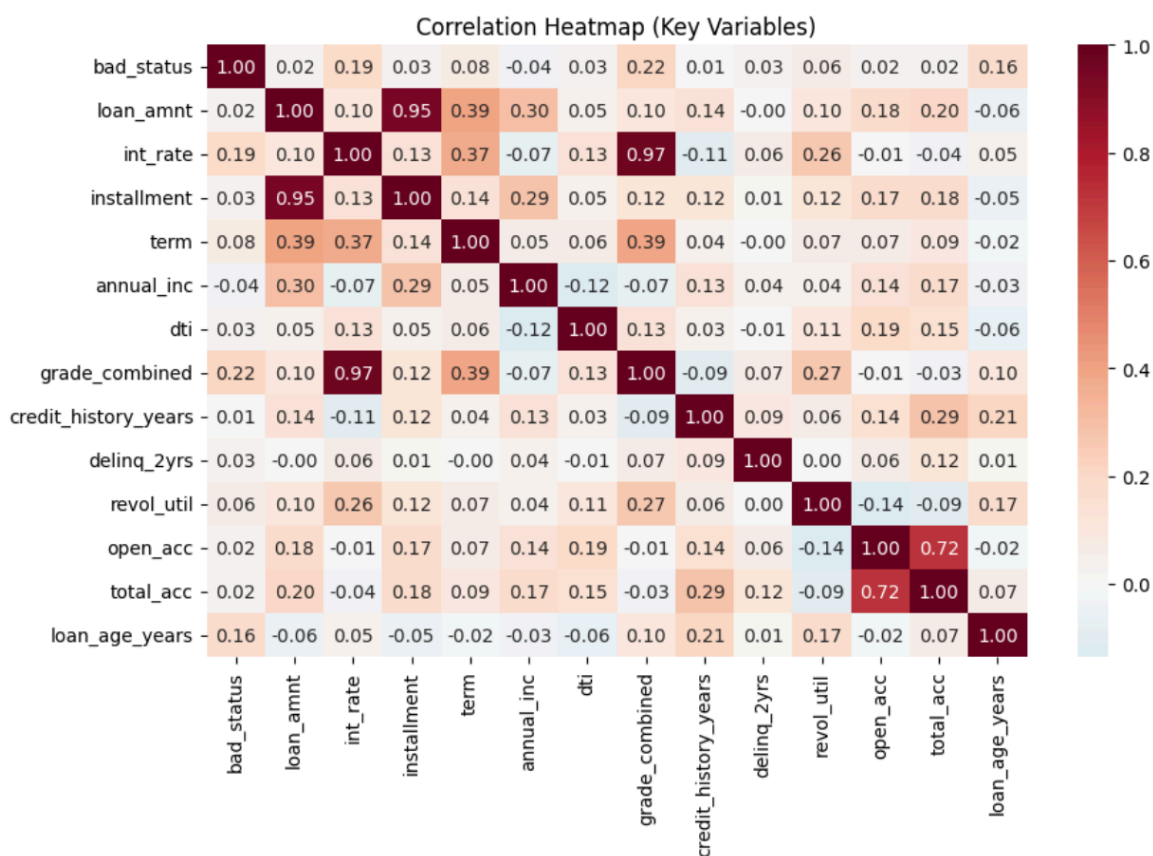
Equity-market and WTI returns generally showed weaker correlations with the one-month-ahead target. More broadly, correlations tended to decline as the lag horizon increased, suggesting that more recent information was generally more closely associated with near-term credit-spread levels.

Importantly, these correlations were treated as descriptive evidence rather than a final feature-selection rule. A strong individual correlation with the target does not necessarily imply that a variable will improve out-of-sample forecasting once it is combined with other predictors. Conversely, a variable with a relatively weak individual correlation may still provide incremental information when used alongside other features. The predictive usefulness of the candidate variables was therefore evaluated subsequently through the forecasting models and out-of-sample performance.

7.2 Correlation Analysis and Multicollinearity

A comprehensive correlation analysis was performed across borrower financial metrics, credit-quality indicators, and derived time-based features. The results showed that **grade_combined** and **interest_rate** had the strongest positive correlations with default, indicating that higher interest rates and lower credit grades are primary indicators of risk.

Furthermore, the analysis identified significant intercorrelations among several independent variables, notably between loan amount and monthly installments, and between credit grade and interest rate. These findings suggest the presence of multicollinearity, which must be accounted for when interpreting the coefficients of linear models such as logistic regression.



9. Modeling and Forecast Evaluation

Following the exploratory analysis and construction of lagged predictors, the next stage of the project develops and evaluates models for forecasting the corporate credit spread one month ahead. The objective is not only to fit the historical relationship between credit spreads and the selected economic and financial indicators, but also to determine whether these relationships provide useful predictive information when forecasting previously unseen periods.

Because the data are time-series observations, model evaluation is designed to preserve their chronological structure and avoid the use of future information when generating forecasts. The analysis therefore uses a time-based training and test split, with the most recent observations reserved for out-of-sample evaluation. For the multivariate forecasting models, an expanding-window approach is subsequently used to simulate how forecasts could have been generated sequentially as new information became available over time.

Model performance is evaluated relative to simple forecasting benchmarks before progressively introducing economic, financial, and lagged credit-spread predictors. Forecast accuracy is assessed primarily using mean absolute error (MAE) and root mean squared error (RMSE), while additional analysis later evaluates the models' ability to correctly predict the direction of credit-spread movements. Statistical tests are also used to assess whether observed forecasting improvements and directional performance are statistically meaningful.

9.1 Time-Based Train-Test Split

Because the objective of this project is to forecast credit spreads one month ahead, the dataset was divided chronologically rather than through a random train-test split. Random splitting would allow observations from later periods to enter the training sample while earlier observations appear in the test sample, which would not reflect how a forecasting model would be used in practice.

The available modeling sample was therefore divided into a training/development period and a strictly later out-of-sample test period. The training sample contains observations from February 1991 through December 2021, while the out-of-sample evaluation period begins in January 2022. This resulted in 371 training observations and 55 test observations.

For the OLS forecasting models evaluated after the benchmarks, an expanding-window approach was used. The model was initially estimated using all observations available before the beginning of the test period. **After each one-month-ahead forecast, the newly observed data became available for**

subsequent model estimation, causing the training window to expand through time. Thus, each forecast was generated using only information that would have been available at that point in the historical sequence.

This approach more closely approximates a real forecasting environment: the model does not have access to future observations, but it can incorporate newly available information as time progresses. Unlike a rolling window, which maintains a fixed sample length by dropping older observations, an expanding window retains the full historical sample and adds new observations as they become available.

9.2 Forecasting Benchmarks

Before evaluating the multivariate regression models, two simple forecasting benchmarks were established. These provide reference points for determining whether the inclusion of macroeconomic, interest-rate, and financial-market information produces meaningful improvements in one-month-ahead credit-spread forecasts.

The first benchmark is a naive persistence forecast, which assumes that next month's credit spread will equal the current month's spread. The second is an autoregressive AR(1) benchmark, which estimates the historical relationship between the current credit spread and the following month's spread. Both are particularly relevant because the earlier time-series analysis demonstrated substantial persistence in credit spreads.

9.2.1 Naive Persistence Forecast

The naive persistence benchmark assumes that the credit spread in the following month will be equal to the credit spread observed in the current month:

$$\text{Predicted Spread}(t+1) = \text{Spread}(t)$$

Despite its simplicity, the naive forecast represents a meaningful benchmark because credit spreads are highly persistent and frequently change gradually from one month to the next. A more complex forecasting model should therefore demonstrate whether it can improve upon the information already contained in the current spread.

Over the out-of-sample test period, the naive persistence forecast produced an **MAE of 0.0931** and an **RMSE of 0.1184**, and an **R² of 0.7666**. Since the credit spread is measured in percentage points, these values correspond to an average absolute forecast error of approximately **9.3 basis points** and an RMSE

of approximately **11.8 basis points**. The higher RMSE reflects the greater weight that this metric places on larger forecast errors.

9.2.2 Autoregressive AR(1) Benchmark

The second benchmark was an autoregressive model with one lag, or AR(1), which uses the current credit spread to forecast the credit spread in the following month:

$$\text{Spread}(t+1) = \text{Intercept} + \beta \times \text{Spread}(t) + \text{Error}$$

Unlike the naive persistence forecast, which assumes that next month's spread will simply equal the current spread, the AR(1) model estimates the historical relationship between consecutive monthly credit-spread observations.

The estimated **coefficient** on the current credit spread was approximately **0.97**, indicating a **very high degree of persistence** in the series and supporting the patterns previously observed in the ACF and PACF analysis. A coefficient close to 1 indicates that the current spread contains substantial information about the level of the spread in the following month.

Over the out-of-sample test period, the AR(1) benchmark produced an **MAE of 0.0948** and an **RMSE of 0.1185**, and an **R² of 0.7664**. Its performance was therefore very similar to those of the naive persistence benchmark, indicating that estimating an autoregressive relationship did not improve forecast performance relative to simply using the current spread as the next-month forecast.

Together, the naive persistence and AR(1) models establish simple but challenging benchmarks for the subsequent multivariate regression models. Because the credit spread is highly persistent, the later models must determine whether macroeconomic, interest-rate, and financial-market information can provide additional predictive value beyond the information already contained in the current credit spread.

9.3 Linear Regression Models

Following the benchmark analysis, a series of ordinary least squares (OLS) regression models were evaluated to determine whether macroeconomic and financial-market information could improve one-month-ahead credit-spread forecasts. The models were developed progressively, beginning with economic and market predictors alone and then incorporating information from the credit spread itself. Multicollinearity among the predictors was also examined before current and lagged variables were combined in the later specifications.

9.3.1 Baseline OLS with Contemporaneous Economic and Market Predictors

The first OLS specification was designed to test **whether contemporaneous macroeconomic and financial-market variables alone could forecast the following month's credit spread**. Importantly, this model excluded both the current credit spread and the six-month rolling credit-spread volatility measure. This allowed the predictive contribution of the external economic and market variables to be evaluated independently of information derived directly from the credit-spread series.

The model performed poorly during the test period, producing an **MAE of 0.3005**, an **RMSE of 0.3588**, and an **R^2 of -1.1418**. These errors were substantially larger than those of both forecasting benchmarks. The negative R^2 further indicates that the model's predictions performed worse, in terms of squared error, than using the test-period mean of the target as a reference prediction.

These results suggest that contemporaneous macroeconomic and market variables alone were insufficient to capture the strong persistence in the monthly credit-spread series.

9.3.2 OLS with Credit-Spread and Economic and Market Predictors

The second OLS specification added the current credit spread and six-month rolling credit-spread volatility to the contemporaneous economic and financial-market predictors. This allowed the model to combine information about the recent state of the credit market with broader macroeconomic and market conditions.

Performance improved substantially. The model achieved an **MAE of 0.0854**, an **RMSE of 0.1057**, and an **R^2 of approximately 0.814**. Both MAE and RMSE were lower than those of the naive persistence and AR(1) benchmarks, while R^2 also increased.

The large improvement relative to the first OLS specification reinforces the importance of the credit spread's own recent behavior in forecasting its next-month level. At the same time, the improvement relative to the persistence-based benchmarks suggests that combining credit-spread information with broader economic and market predictors may provide additional predictive information.

9.3.3 Multicollinearity Diagnostic

Because several economic and interest-rate predictors were correlated, multicollinearity was examined using the variance inflation factor (VIF). The diagnostic identified particularly elevated VIF values among interest-rate variables, including the federal funds rate, 10-year Treasury yield, and 10-year minus 3-month Treasury yield spread. This was consistent with relationships observed earlier in the correlation analysis.

To examine whether reducing this multicollinearity improved the model, the 10-year Treasury yield was removed and the regression was re-estimated. Although the VIF values declined, out-of-sample forecasting performance deteriorated: MAE and RMSE increased and R^2 fell to approximately 0.63.

Because the primary objective of the project is forecasting rather than interpreting individual regression coefficients, the correlated predictors were retained in the subsequent analysis. While multicollinearity can make individual OLS coefficients less stable and more difficult to interpret, removing the 10-year Treasury yield reduced out-of-sample predictive performance in this case. The final feature-selection decisions therefore considered forecasting performance in addition to the multicollinearity diagnostic.

9.3.4 OLS with Selected Current and Lagged Predictors

The final OLS specification examined whether combining current credit-spread information with selected current and lagged macro-financial predictors could further improve one-month-ahead forecasting performance. As described in Section 9.1, **the model was evaluated using an expanding-window procedure.**

Predictor selection was informed by the earlier correlation and lag analysis, with greater consideration given to variables that showed stronger relationships with the forecast target. Several alternative combinations of current and lagged predictors were also evaluated during model development, and the final specification was selected based on out-of-sample forecasting performance.

The final predictor set included:

- Credit-spread dynamics: current credit spread, one-month lagged credit spread, and six-month rolling credit-spread volatility
- Market risk: VIX
- Monetary policy: federal funds rate
- Interest-rate conditions: current 10-year Treasury yield, one-month lagged 10-year Treasury yield, and 10-year minus 3-month Treasury yield spread

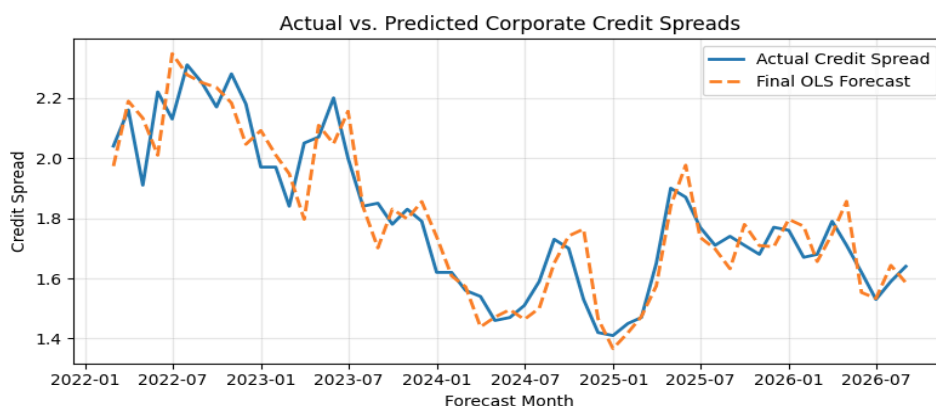
- Equity markets: current S&P 500 monthly return and its three-month lag; one-month and three-month lagged Russell 2000 returns
- Macroeconomic conditions: year-over-year inflation, unemployment rate, industrial production growth, and real GDP growth
- Energy market: three-month lagged WTI monthly return

The final model achieved an **MAE of 0.0768**, an **RMSE of 0.0993**, and an **R² of 0.8359**, representing the strongest overall forecasting performance among the models evaluated. Relative to the naive persistence benchmark, the model **reduced MAE by approximately 17.5%** and **RMSE by approximately 16.1%**, while **R² increased from 0.7666 to 0.8359**. The model also outperformed the AR(1) benchmark and the earlier OLS specification using the current credit spread, rolling volatility, and contemporaneous economic and market predictors.

These results suggest that incorporating selected lagged macro-financial information alongside current and recent credit-spread dynamics provides additional predictive value for one-month-ahead credit-spread forecasting. The actual-versus-predicted results show that the model generally tracked the realized evolution of credit spreads during the test period, although forecast errors were more pronounced around sharper month-to-month movements.

Out-of-Sample Model Performance Comparison

	Model	MAE	RMSE	R ²
0	Naive Persistence	0.0931	0.1184	0.7666
1	AR(1)	0.0948	0.1185	0.7664
2	OLS – Spread + Economic/Market Predictors	0.0854	0.1057	0.8140
3	Final Expanding-Window OLS	0.0768	0.0993	0.8359



The actual-versus-predicted plot shows that the final OLS forecasts generally track the evolution of realized credit spreads over the test period and capture several turning points in the series. Forecast errors are more pronounced around sharper month-to-month movements, with the model occasionally overestimating the magnitude of spread changes. Overall, the model captures the broader movement in credit-spread levels while remaining less precise in forecasting the magnitude of sudden changes.

10. Directional Forecast Evaluation

Forecast-error measures such as MAE and RMSE evaluate the magnitude of prediction errors but do not indicate whether a model correctly anticipates the direction of credit-spread movements. From an investment and risk-management perspective, the ability to distinguish between spread widening and tightening may provide useful information even when the exact magnitude of the subsequent spread is difficult to forecast. Directional performance was therefore evaluated separately for the AR(1) benchmark and the final OLS model.

For each out-of-sample observation, **the actual change** was calculated as the next month's actual credit spread minus the current month's credit spread. **The predicted change** was calculated as the model's forecast for the next month minus the current month's credit spread.

A positive change represents **spread widening**, while a negative change represents **spread tightening**. Directional accuracy was then calculated as the percentage of test observations for which the predicted direction matched the actual direction

$$\text{Actual Change}_t = \text{Spread}_t - \text{Spread}_{t-1} \quad \text{Predicted Change}_t = \widehat{\text{Spread}}_t - \text{Spread}_{t-1}$$

$$\text{Directional Accuracy} = \frac{\text{Number of Correct Direction Predictions}}{\text{Total Number of Forecasts}} \times 100$$

10.1 Directional Accuracy

The AR(1) benchmark correctly predicted the direction of **24 of the 55 out-of-sample observations**, producing a directional **accuracy of 43.64%**. However, examination of the individual directional forecasts reveals **an important limitation of this result**. The AR(1) model **predicted spread widening in all 55 test observations** and never predicted tightening. The actual sample contained 24 widening observations, 29 tightening observations, and 2 observations with no change.

Consequently, the AR(1)'s 43.64% directional accuracy should not be interpreted as evidence of meaningful directional forecasting ability. Its 24 correct predictions occurred entirely because 24 months in the test sample happened to experience widening. The result therefore depends heavily on the realized frequency of widening and tightening during the selected test period rather than on the model successfully distinguishing between the two directions.

The final OLS model produced substantially more balanced directional forecasts. **It predicted 26 months of widening and 29 months of tightening**, compared with 24 actual widening months and 29 actual tightening months, with 2 months exhibiting no change. Overall, the model correctly predicted **36 of 55 observations**, corresponding to a **directional accuracy of 65.45%**. Unlike the AR(1) benchmark, the final OLS model generated forecasts in both directions, indicating an ability to distinguish between periods of predicted spread widening and tightening.

Model	DA	Predicted Tightening	Predicted Widening	Predicted Unchanged
AR(1)	43.64%	0	55	0
Final OLS	65.45%	29	26	0
Actual	—	29 Tightening	24 Widening	2 Unchanged

10.2 Statistical Significance of Directional Forecasts: Pesaran–Timmermann Test

Directional accuracy alone does not establish that a model contains statistically meaningful directional forecasting information. A model could achieve apparent directional accuracy simply because one direction occurs more frequently than another or because the model itself disproportionately predicts one direction. To evaluate whether the directional agreement produced by the final OLS model exceeds what would be expected from these underlying frequencies, the **Pesaran–Timmermann (PT) test was applied**.

The PT test evaluates whether the predicted and actual directions of change are statistically associated. **The null hypothesis is that predicted and actual spread directions are independent**, meaning that any observed directional agreement can be explained by the frequencies with which widening and tightening occur rather than by genuine directional forecasting ability. **The alternative hypothesis is that predicted and actual directions are associated**.

The two observations with no actual change were excluded from the binary widening-versus-tightening PT test, leaving 53 observations. Among these observations, the actual sample contained 24 widening and 29 tightening months, while the model predicted 25 widening and 28 tightening months. The expected directional agreement under independence was approximately **50.27%**.

The Pesaran–Timmermann test produced a **PT statistic of 2.5866** and a two-sided **p-value of 0.0097**. Because the **p-value is below 0.01**, the **null hypothesis of independence is rejected at the 1% significance level**. This provides evidence that the final OLS model's directional forecasts contain statistically significant information about subsequent credit-spread movements rather than its 65.45% directional accuracy being explained solely by the relative frequencies of widening and tightening in the test sample.

11. Statistical Comparison of Forecast Accuracy: Diebold–Mariano Test

Although MAE and RMSE showed that the final OLS model produced smaller out-of-sample forecast errors than the benchmark models, lower error metrics alone do not establish whether the improvement in forecast accuracy is statistically significant. The **Diebold–Mariano (DM) test** was therefore used to compare the predictive accuracy of the final expanding-window OLS model with the **naive persistence benchmark**, which was the stronger of the two benchmark models based on MAE and RMSE.

The test compares the forecast losses produced by the two models for each of the **55 out-of-sample observations**. Squared forecast errors were used as the loss function, consistent with the RMSE criterion used in the model evaluation. For each month, the squared error of the naive forecast was subtracted from the squared error of the final OLS forecast. The average of these monthly loss differences indicates whether one model tended to produce smaller forecast errors over the test period.

The hypotheses of the DM test are:

Null hypothesis: The final OLS model and the naive persistence benchmark have equal predictive accuracy.

Alternative hypothesis: The two models have different predictive accuracy.

The average loss difference was -0.004163 . Because the loss difference was defined as the squared error of the OLS model minus the squared error of the naive model, the negative value indicates that the final OLS model produced lower squared forecast loss on average.

The Diebold–Mariano test produced a **DM statistic of -1.8142** and a **p-value of 0.0697** . At the conventional **5% significance level**, the null hypothesis of equal predictive accuracy **cannot be rejected**. Therefore, although the final OLS model produced lower MAE and RMSE, the evidence is not sufficiently strong to conclude at the 5% level that its forecast accuracy is statistically different from that of the naive benchmark.

However, the **result is statistically significant at the 10% level**, providing moderate evidence that the final OLS model improves predictive accuracy relative to the naive persistence benchmark. The relatively small out-of-sample test period of 55 monthly observations may also limit the statistical power of the comparison. Overall, the DM results provide some statistical support for the improvement observed in the forecast-error metrics, although the evidence is weaker than the directional significance established by the Pesaran–Timmermann test.

11. Conclusion

This project examined whether macroeconomic and financial indicators can improve one-month-ahead forecasts of corporate credit spreads beyond simple persistence and autoregressive benchmarks. Several model specifications were evaluated, with **the final expanding-window OLS model incorporating selected current and lagged credit-spread and macro-financial predictors providing the strongest overall out-of-sample performance**.

The final OLS model achieved an **MAE of 0.0768** , **RMSE of 0.0993** , and **R^2 of 0.8359** , outperforming both the naive persistence and AR(1) benchmarks on the point-forecast metrics. Relative to the naive persistence benchmark, this represents reductions of approximately **17.5% in MAE** and **16.1% in RMSE**.

The model also demonstrated substantially **stronger directional performance**. It correctly predicted whether credit spreads would widen or tighten in **36 of 55 months**, corresponding to a **directional accuracy of 65.45%**, compared with 43.64% for the AR(1) benchmark. Importantly, the AR(1) model predicted spread widening in every test observation, meaning that its directional accuracy reflected the frequency of actual widening observations rather than an ability to distinguish between widening and tightening. In contrast, the final OLS model predicted both directions and produced a substantially more balanced set of directional forecasts.

The Pesaran–Timmermann test provided evidence that the final model’s directional predictive ability was statistically significant, producing a p-value of 0.0097 . The null hypothesis of independence between predicted and actual directions was therefore rejected at the 1% significance level, providing strong evidence that the model contained meaningful information about the direction of subsequent credit-spread movements.

The Diebold–Mariano test, which compared squared forecast losses between the final OLS model and the naive persistence benchmark, produced a **DM statistic of -1.8142** and a **p-value of 0.0697** . The negative statistic was consistent with the lower average squared forecast loss of the final OLS model. Although the null hypothesis of equal predictive accuracy **could not be rejected at the conventional 5% significance level, it could be rejected at the 10% level**, providing moderate statistical evidence that the improvement in point-forecast accuracy was not solely due to sampling variation.

Overall, the results suggest that combining recent credit-spread dynamics with selected macroeconomic and financial-market information provides useful incremental information for one-month-ahead credit-spread forecasting. The strongest evidence was found for predicting the direction of future spread movements, while the model also produced economically meaningful improvements in the magnitude of its point forecasts relative to simple benchmarks.

12. Limitations and Future Work

Several limitations of the analysis provide opportunities for extending the forecasting framework.

Data availability and temporal alignment represented one of the main constraints of the project.

The economic and financial indicators considered in the analysis differed substantially in historical coverage, frequency, and publication schedules. Some potentially useful predictors could not be incorporated because sufficiently long historical or release-aware data were unavailable. For example, measures based on economic surprises or expectations could potentially provide more forward-looking information, but comparable histories were not available over the full modeling period. The final sample therefore begins in the early 1990s, allowing the project to preserve a reasonably broad and consistent set of predictors.

Temporal alignment also required conservative assumptions about when information became available. Macroeconomic indicators are released according to different schedules, and some become available considerably later than others. To reduce the risk of look-ahead bias, predictors were aligned according to their historical availability before being used for forecasting. While this creates a more realistic forecasting exercise, conservative alignment can make some economic information effectively more lagged. Future work could improve this process through more comprehensive point-in-time or vintage databases and could investigate forecasting designs that more explicitly account for differences in economic release dates.

A second limitation is the relatively small out-of-sample test period of 55 monthly observations.

Although this provides a meaningful recent forecasting exercise, the limited number of observations reduces the statistical power of forecast-comparison tests and increases uncertainty around estimated performance differences. This is particularly relevant to the Diebold–Mariano test, where the final model produced lower forecast errors but the difference was statistically significant at the 10% level rather than the conventional 5% level. Future research could evaluate the framework over a longer out-of-sample period and across multiple economic and credit-market regimes.

A third limitation concerns model and feature selection. Several model specifications, combinations of current and lagged predictors, and alternative feature sets were explored during model development. However, the analysis did not use separate training, validation, and final holdout periods. A more rigorous extension could perform feature and model selection exclusively within a validation period before evaluating the selected specification once on a completely untouched test set. Lag selection could also be conducted more systematically across candidate predictors and forecasting horizons.

Finally, several predictors exhibited substantial correlation, particularly among related interest-rate, equity-market, and macroeconomic variables. **Multicollinearity was examined during model development, and alternative specifications and regularized regression were explored. Ridge regression produced results similar to OLS, suggesting that regularization provided limited additional benefit for the specifications considered.** However, a more systematic variable-selection procedure could evaluate correlated predictors individually and in combination and examine their stability across different forecasting periods. This would be particularly important if the objective were to interpret individual regression coefficients rather than primarily evaluate forecasting performance.

Future work could therefore focus on building a richer and longer release-aware dataset, incorporating additional forward-looking indicators, performing more systematic feature and lag selection, and evaluating the forecasting framework over a larger independent test sample and across different economic and credit-market regimes.

